

Using Cisco Packet Tracer Hotel Network Configuration

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Abstract: This paper presents the design and implementation of a secure and scalable hotel network using Cisco Packet Tracer. The configuration leverages Virtual Local Area Networks (VLANs) for efficient traffic management, inter-VLAN routing for seamless communication, and Dynamic Host Configuration Protocol (DHCP) for automated IP address allocation. Strong security protocols are implemented to ensure network reliability. This network is tailored to meet the operational requirements of modern hotels, supporting administrative functions and guest services while ensuring high performance and security.

Keywords— *Hotel Network, VLAN, Inter-VLAN Routing, DHCP, Network Security, Cisco Packet Tracer, Scalable Infrastructure.*

I. INTRODUCTION

The hospitality industry has increasingly relied on sophisticated information technology (IT) infrastructures to enhance service delivery and streamline operations. A well-structured network allows hotels to manage guest requests, control room access, and facilitate inter-departmental communication effectively. This project details the design and configuration of a hotel network using Cisco Packet Tracer, a powerful simulation tool that enables the modeling of complex network setups.

This project focuses on creating a network that addresses the immediate needs of the hotel and allows for future expansion and integration of new technologies. By implementing VLANs, we can segment traffic by department, which enhances security and improves network performance. This paper discusses the objectives, design methodologies, configurations, and expected outcomes of the hotel network project.

II. OBJECTIVES

The primary objectives of this project are as follows:

1. **Design a Scalable Network:** Develop an infrastructure capable of accommodating growth in the number of users and devices.
2. **Implement VLAN Segmentation:** Manage departmental traffic effectively and enhance security by isolating different network segments.
3. **Configure Routers for Inter-VLAN Communication:** Enable seamless communication between different VLANs.
4. **Deploy DHCP for Efficient IP Management:** Automate the allocation of IP addresses to devices within the network.

5. **Establish Strong Security Measures:**
Protect sensitive data and network integrity through robust security protocols.

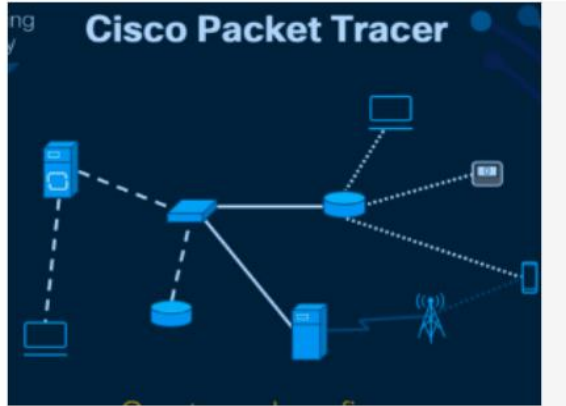


Fig. 1: Cisco Packet Tracer interface

III. METHODOLOGY

The methodology for this project involved several structured steps:

1. Requirement Analysis:

- Assess the operational needs of various departments, including administration, reception, marketing, IT, and guest services.
- Determine the types and numbers of devices that will connect to the network, including computers, tablets, smartphones, and printers.

2. Network Design:

- Create a detailed network topology that incorporates VLANs and routing devices.
- Define the IP addressing scheme, ensuring that each VLAN has a unique subnet to facilitate organization and management.

3. Configuration:

- Set up VLANs to segregate traffic by department, configure routers for inter-VLAN

communication, and deploy DHCP to automate IP address allocation.

- Use Cisco Packet Tracer to simulate the network environment, allowing for real-time testing and adjustments.

4. Testing:

- Conduct simulations to verify connectivity between devices in different VLANs and ensure that DHCP is allocating IP addresses correctly.
- Test security configurations by attempting unauthorized access to ensure that ACLs and password protections are effective.

5. Documentation:

- Compile results and observations throughout the testing process to create a comprehensive report detailing the design and functionality of the network.

IV. NETWORK DESIGN

The network is divided into several VLANs, each corresponding to a specific department within the hotel, allowing for efficient management and security:

- **Admin Office VLAN (192.168.10.0):**
Dedicated to administrative tasks and management systems, including computers and printers used by hotel management.
- **Reception VLAN (192.168.12.0):**
Supports guest-facing operations, including check-in/out terminals and communication devices.
- **Marketing VLAN (192.168.13.0):**
Isolates marketing communications, allowing the marketing team to manage campaigns and communications without interference.

- **IT Department VLAN (192.168.19.0):**
Provides technical support and IT operations, ensuring a secure environment for sensitive data, such as guest information and financial records.
- **Guest Floors VLANs (192.168.15.0 & 192.168.18.0):**
Facilitate guest services, allowing connections for guest devices (smartphones, laptops) and internal staff communication for room service and maintenance requests.

The topology is designed to include routers to manage inter-VLAN traffic and switches to connect devices within each VLAN.

Serial Router Configuration

Router 1

```
Router1
Physical Config CLI Attributes
IOS Command Line Interface

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip add 192.168.11.1 255.255.255.0
Router(config-if)#clock rate 56000
Router(config-if)#no sh

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to down
Router(config-if)#int fa0/0
Router(config-if)#ip add 192.168.10.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/3/0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 0.0.0.0 0.0.0.0 s0/1/0
%Default route without gateway, if not a point-to-point interface, may impact performance
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
```

Router 2

```
Router2
Physical Config CLI Attributes
IOS Command Line Interface

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface serial 0/3/0
Router(config-if)#ip add 192.168.11.2 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
Router(config-if)#interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up
Router(config-if)#exit
Router(config)#interface serial 0/1/0
Router(config-if)#ip add 192.168.14.1 255.255.255.0
Router(config-if)#no sh

%LINK-5-CHANGED: Interface Serial0/1/0, changed state to down
Router(config-if)#int fa0/0
Router(config-if)#ip add 192.168.16.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#int fa0/1
Router(config-if)#ip add 192.168.15.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Router 3

```
Router3
Physical Config CLI Attributes
IOS Command Line Interface

Press RETURN to get started!

Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface s0/1/0
Router(config-if)#ip add 192.168.14.2 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/1/0, changed state to up
Router(config-if)#e
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/1/0, changed state to up
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int fa0/0
Router(config-if)#ip add 192.168.18.1 255.255.255.0
Router(config-if)#no sh

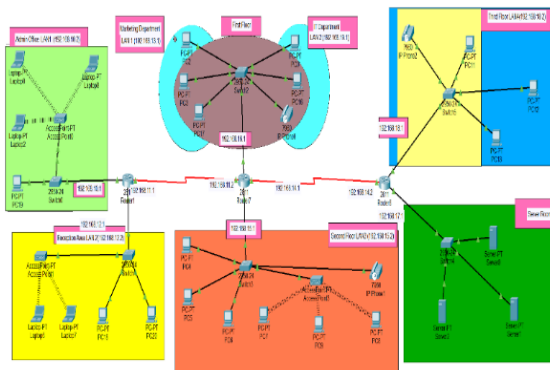
Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up
Router(config-if)#int fa0/1
Router(config-if)#ip add 192.168.17.1 255.255.255.0
Router(config-if)#exit
Router(config)#int fa0/1
Router(config-if)#ip add 192.168.17.1 255.255.255.0
Router(config-if)#no sh

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
Router(config-if)#exit
Router#
```

V. NETWORK TOPOLOGY

The network topology comprises the following components:

1. Routers: Facilitate interconnection between VLANs and provide gateway access to the internet.
2. Switches: Serve to connect devices within each VLAN segment.
3. PCs: Representing guest devices including laptops, smartphones.
4. Web Server: Hosts essential services such as guest portal, room service requests, etc. DHCP Server: Facilitates automatic IP address assignment to network devices.



Configurations

Router Configurations

- **Static IP Assignment:** Each VLAN is assigned a unique IP address to serve as the gateway for routing traffic.
- **Inter-VLAN Routing:** Routers are configured with sub-interfaces for each VLAN, allowing devices in different VLANs to communicate. For example, Router1 may have sub-interfaces configured for

- 192.168.10.1 (Admin) and 192.168.12.1 (Reception).

DHCP Setup

- **Dynamic IP Allocation:** A DHCP server is set up to automatically assign IP addresses to devices as they connect to the network. This reduces administrative overhead and ensures that IP address management is efficient. DHCP pools are defined for each VLAN to ensure that IP addresses are allocated from the correct subnet.

Switch Configurations

VLAN Implementation: Each switch is configured with VLANs corresponding to departmental needs, facilitating traffic detailing the design and functionality of the network

1. Stick VLAN Implementation Switch

```
10 marketing          active
20 it                 active
1002 fddi-default      active
1003 token-ring-default active
1004 fddinet-default   active
1005 trnet-default     active
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range f0/2-3
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#int range f0/4-6
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#show vlan br
```

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```
10 marketing          active
20 it                 active
1002 fddi-default      active
1003 token-ring-default active
1004 fddinet-default   active
1005 trnet-default     active
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int range f0/2-3
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#exit
Switch(config)#int range f0/4-6
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr
Building configuration...
[OK]
Switch#show vlan br
```

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☐ Top

2. Password Setting

Admin Office

Access Point0

Physical Config Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port Status

SSID Admin

2.4 GHz Channel 6

Coverage Range (meters) 140.00

Authentication

☐ Disabled ☐ WEP WEP Key

☐ WPA-PSK ☒ WPA2-PSK PSK Pass Phrase adminoffice

User ID

Password

Encryption Type AES

Second Floor

Access Point3

Physical Config Attributes

GLOBAL

Settings

INTERFACE

Port 0

Port 1

Port Status

SSID Second

2.4 GHz Channel 6

Coverage Range (meters) 140.00

Authentication

☐ Disabled ☐ WEP WEP Key

☐ WPA-PSK ☒ WPA2-PSK PSK Pass Phrase secondfloor

User ID

Password

Encryption Type AES

3. DHCP Implementation

Router 1

```
Router#en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool lan1
Router(dhcp-config)#default-router 192.168.10.1
Router(dhcp-config)#network 192.168.10.2 255.255.255.0
Router(dhcp-config)#exit
Router(config)#
```

Copy Paste

Router 2

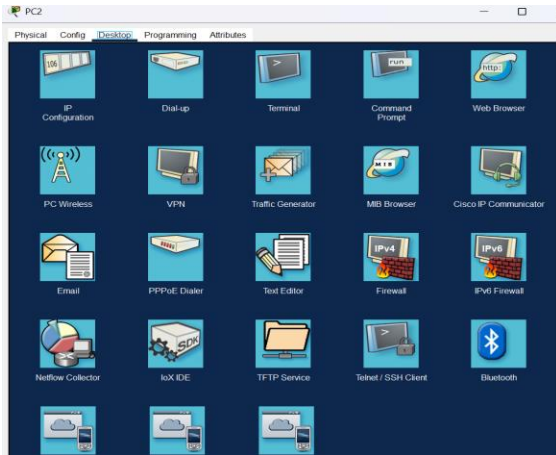
```
Router#en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool lan2
Router(dhcp-config)#default-router 192.168.16.1
Router(dhcp-config)#network 192.168.16.2 255.255.255.0
Router(dhcp-config)#exit
Router(config)#ip dhcp pool lan3
Router(dhcp-config)#default-router 192.168.15.1
Router(dhcp-config)#network 192.168.15.2 255.255.255.0
Router(dhcp-config)#exit
```

Router 3

```
Router#en
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip dhcp pool lan4
Router(dhcp-config)#default-router 192.168.18.1
Router(dhcp-config)#network 192.168.18.2 255.255.255.0
Router(dhcp-config)#exit
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Security Measures

- **Password Protection:** Strong passwords are enforced on all network devices to prevent unauthorized access. Default passwords are changed during initial configuration.
- **Access Control Lists (ACLs):** ACLs are implemented to control traffic flow between VLANs, ensuring that only authorized users can access sensitive areas of the network. For instance, only IT personnel may access the IT VLAN.
- **Physical Security:** Critical network components, such as servers and routers, are located in secure areas to prevent tampering. Access to these areas is limited to authorized personnel only.



VI. CONCLUSION

This project successfully demonstrates the design and implementation of a secure and efficient hotel network using Cisco Packet Tracer. The incorporation of VLANs, inter-VLAN routing, and DHCP has resulted in a network that meets the specific needs of hotel operations while providing scalability and security.

VII. FUTURE Work

Future enhancements could focus on:

- **Integration of IoT Devices:** Incorporating smart technology to enhance guest services, such as smart room controls, energy management systems, and automated check-in processes.
- **Expanded Wireless Coverage:** Ensuring that all areas of the hotel have robust wireless connectivity, potentially through additional access points and load balancing techniques.
- **Real-Time Monitoring:** Implementing network monitoring tools to proactively manage and address potential issues. This could include monitoring

bandwidth usage, device connectivity, and security breaches.

VIII. REFERENCES

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